
Persistent Data-Order Effects Precede Directional Alignment in Language Models

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Abstract

Data order changes optimization trajectories, but endpoint divergence alone does not reveal how the corresponding trace develops or whether it is recoverable. We measure output divergence and raw parameter-space endpoint-direction alignment along public Pythia-160M/PolyPythias trajectories trained on 300 billion tokens. Three correlated order-only pairs reach symmetric KL of 0.290–0.293 nats, approximately 3.8 times a late-training adjacent-checkpoint drift reference, compared with 0.306 nats for one joint initialization-and-order pair. The central finding is a temporal dissociation: at step 16,000, the order/joint divergence ratio is already 0.932 for one pair, while its raw endpoint-direction cosine is only 0.294. These are distinct readouts, not a causal decomposition. Most parameter-block grafts then leave a predeclared language-modeling loss band, and none of the tested Fisher estimators predicts all three probe directions within 25% relative error. The comparisons use one training-distribution shard and lack an initialization-only control. Persistent order dependence is measurable; functional recoverability and the cost of rewriting completed optimization histories remain open.

1 Introduction

Training is an ordered composition of update maps: identical initial states and data multisets can yield different completed histories. Endpoint distance leaves several questions separate. Does an output-level difference persist? When does its magnitude become substantial, and when does the raw parameter-space endpoint-difference direction align with its final direction? Can that trace be corrected, and can a proposed recovery probe be trusted?

A scalar output divergence need not track parameter orientation: later dynamics may change a displacement’s direction while preserving the scale of its effect on predictions. We examine this distinction using public Pythia and PolyPythias trajectories [1, 2], without new pretraining. Our contributions are:

1. **Temporal dissociation.** Functional divergence is near its late-training scale while the raw endpoint-difference direction remains weakly aligned: at step 16,000, the order/joint ratio is 0.932 and the cosine is 0.294 for the primary pair. Qualitative agreement across correlated pairs has pair-dependent timing.
2. **Persistent output differences.** Three correlated order-only comparisons reach 0.290–0.293 nats at the final Pythia-160M checkpoint. These are pairwise comparisons sharing one initialization, not independent replicates.

3. **Validity-gated recovery analysis.** The tested graft class and Fisher estimators fail declared validity criteria. Their failure prevents recovery claims, rather than establishing irreversibility.

2 Dynamical framework and measurement design

Transported order defects. Let x include parameters and any optimizer state, and let U_i be one update map. The adjacent-swap defect is

$$\delta_{ij}(x) = U_j(U_i(x)) - U_i(U_j(x)). \quad (1)$$

For smooth SGD updates $U_i(x) = x - \eta g_i(x)$, $\delta_{ij}(x) = \eta^2(Dg_j(x)g_i(x) - Dg_i(x)g_j(x)) + O(\eta^3)$. Thus the leading local defect is the established commutator term [5, 6]. Stateful optimizers require the augmented state; the parameter-only SGD expansion cannot be applied unchanged to Adam.

Connect two update words by adjacent swaps. For swap k , let z_k be the state after its first ordering, $z_k + \delta_k$ the state after its alternative, and A_k their common remaining update composition. Their endpoint increment is exactly

$$\Delta_k = A_k(z_k + \delta_k) - A_k(z_k), \quad x_T^{(B)} - x_T^{(A)} = \sum_k \Delta_k. \quad (2)$$

The second identity telescopes over intermediate words with a common initial state and the same update maps. Differentiability gives $\Delta_k \approx DA_k(z_k)\delta_k$: later dynamics can amplify, attenuate, or rotate local defects, which can also cancel in the sum. Checkpoint differences below are direct observations, not reconstructed per-swap attributions. The identity does not identify when Pythia’s individual order effects were generated.

A restricted correction floor. For a parameter gap Δ , fixed linear correction subspace S , and fixed quadratic metric $\Sigma \succeq 0$, define

$$r_S(\Delta) = \min_{c \in S} \|\Sigma^{1/2}(\Delta - c)\|_2. \quad (3)$$

Every correction $c \in S$ has residual norm at least $r_S(\Delta)$, directly by minimization; the minimum is attained since $\Sigma^{1/2}S$ is a finite-dimensional subspace. This elementary conditional floor is neither a nonlinear irreversibility theorem nor a computational lower bound, and does not cover state-dependent reachable spaces. Section 4 tests the metric prerequisite; no functional correction floor is measured here.

Models, metric, and controls. Write m, d_1, d_2 for the main Pythia-160M run and its two data-seed variants. They share an initialization. We analyze $P_1 = (d_1, d_2)$, $P_2 = (m, d_1)$, and $P_3 = (m, d_2)$; these are three correlated pairs from three runs, not three independent replicates. The joint arm J uses the two full-seed variants, changing initialization and order together. The endpoint is $T = 143,000$ updates (300B training tokens).

For next-token distributions on a frozen shard E , our direct readout is

$$D_E(a, b) = \frac{1}{2|E|} \sum_{x \in E} [\text{KL}(p_a(\cdot|x)||p_b(\cdot|x)) + \text{KL}(p_b(\cdot|x)||p_a(\cdot|x))]. \quad (4)$$

The shard contains 48 non-overlapping 2,048-token blocks from Nee1Nanda/pile-10k, concatenated with EOS separators using the Pythia tokenizer: 98,304 prediction positions. It samples the training distribution, not a held-out generalization benchmark. Models run in evaluation mode with float32 forward passes and float64 accumulation. We compare 12 checkpoints from step 0 to T , excluding step 1 after a checkpoint-integrity audit. The candidate initialization-only arm failed continuity checks and was excluded before the amended comparison; $D_E(J) - D_E(P_i)$ confounds initialization and initialization–order interaction. The amended order-only criterion is weaker than the original two-arm criterion.

Positioning. Local Lie-bracket criteria predict order effects in domain and sequential learning [5, 6]; mode-connectivity work studies instability to training noise [4]. Pythia/PolyPythias enable trajectory and seed comparisons [1, 2]. Here the focus is direct output-distribution divergence between completed histories, its timing relative to raw endpoint-direction alignment, and probe validity before recovery claims. We claim neither novelty for order sensitivity itself nor a new general transport theorem.

3 Persistent functional divergence precedes directional alignment

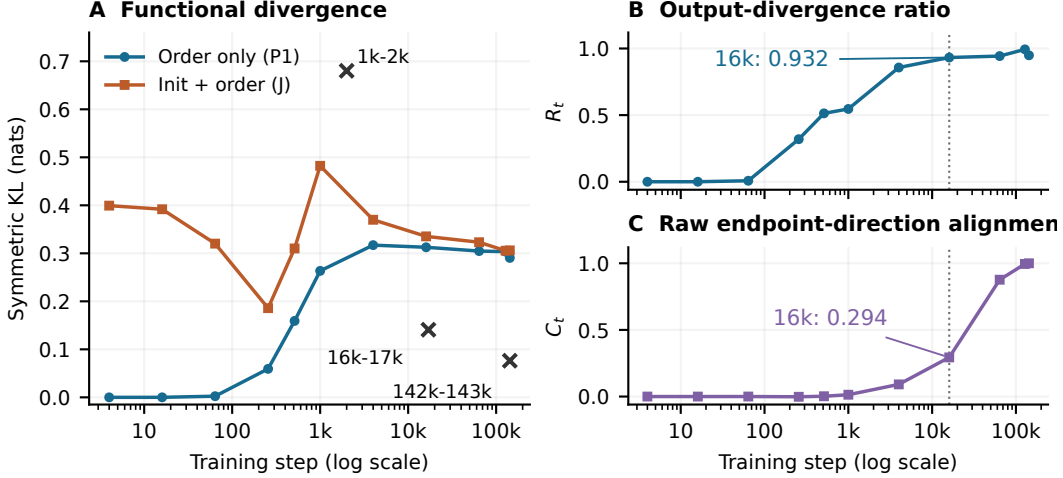


Figure 1: **Substantial functional divergence before raw endpoint-direction alignment.** Left: P_1 and joint-pair output divergence; crosses mark drift over the labeled windows. Right: separate panels show $R_t = D_E(P_1, t)/D_E(J, t)$ and raw cosine C_t . At 16k, $R_t = 0.932$ while $C_t = 0.294$. Ratio and cosine measure different objects; their numerical proximity has no metric interpretation. Lines connect recorded points, not fitted dynamics; crosses are not error bars.

Late-training order effects persist. The final order-only divergences are 0.290–0.293 nats (Table 1), compared with 0.306112 for J . All exceed the late adjacent-checkpoint drift reference, 0.076275, and lie below the 160M–410M comparison, 0.460360. Their maximum/minimum ratio is 1.009, below the predeclared threshold of 2.0. The approximately 1% agreement is descriptive, not a confidence interval: each of three shared-initialization runs appears in two comparisons.

Table 1: Final-checkpoint divergence on the frozen training-distribution shard. All three order-only pairs share one initialization; each run occurs twice. Ratios share the same single joint-pair denominator.

Pair	D_E (nats)	$D_E/D_E(J)$	$D_E/\text{drift reference}$
P_1 : data-seed1 / data-seed2	0.290461	0.9489	3.81
P_2 : main / data-seed1	0.290638	0.9494	3.81
P_3 : main / data-seed2	0.293055	0.9573	3.84
J : full-seed1 / full-seed2	0.306112	1.0000	4.01

The central dissociation: magnitude precedes raw alignment. For P_1 , let $\Delta_t = \theta_{d_1, t} - \theta_{d_2, t}$ and $C_t = \langle \Delta_t, \Delta_T \rangle / (\|\Delta_t\|_2 \|\Delta_T\|_2)$. At 16,000, $D_E(P_1) = 0.312597$ nats, already near its final scale, while $C_t = 0.294$. This statement holds in absolute KL, without the joint denominator. The ratio $R_t = D_E(P_1, t)/D_E(J, t)$ is 0.932. At 64,000, $R_t = 0.943$ and $C_t = 0.877$; the first sampled $C_t \geq 0.9$ occurs at 128,000. The ratio is not monotone: it falls from 0.993 at 128,000 to 0.949 at T .

At 16,000, P_2 and P_3 have raw cosines 0.441 and 0.466, and both first cross 0.9 at 64,000. The same qualitative dissociation occurs in all three correlated pairs, with pair-dependent timing. **Functional divergence magnitude and raw parameter-space endpoint orientation evolve on different timescales.** This is a descriptive comparison, not a test of functional-direction lock-in. Early directions are storage-quantization limited; $C_T = 1$ is an identity and nearby checkpoints correlate mechanically. Threshold-crossing times are grid-dependent, not precise stabilization times.

The drift reference is not a noise floor. It depends on training time and checkpoint interval: over 1,000-update windows it is 0.680225 at 1,000–2,000 and 0.140831 at 16,000–17,000. Early pair values do not establish separation from drift; the 3.8-fold comparison applies only late in training. Neither a scalar ratio nor raw cosine identifies an output-space direction or recoverability.

4 Recovery claims require validated probes

Parameter-type grafts leave the allowed loss region. Raw alignment does not measure recovery. At T , we form hybrids $\theta_h = \theta_{d_1} + \alpha M_b(\theta_{d_2} - \theta_{d_1})$, where M_b selects a parameter-type block and $\alpha \in \{0.25, 0.5, 1\}$. The frozen rule rejects token cross-entropy above the worse parent’s loss plus 0.5 nats: a threshold of 3.001560. Instrument checks passed, including parent-divergence reproduction to relative error 1.4×10^{-10} .

Seven of eight complete block swaps are invalid. The recorded loss-band tally rejects 23 of 35 evaluated graft readouts, including control and second-anchor readouts of the same hybrids. Attention QKV, carrying 87% of the squared parameter gap, is invalid at every tested nonzero strength. Only internal LayerNorm remains valid through a full swap; that hybrid is degraded and far from both parents. The frozen abort rule stops the remaining control sequence, leaving spatial recoverability inconclusive. *Parameter-space graftability cannot be assumed to measure functional recoverability*: this graft class fails at the tested strengths, without ruling out other edits.

Local Fisher metrics do not certify. To attempt Eq. (3) without grafts, we test $D_E(\theta, \theta + v) \approx v^\top \Sigma_E v$, with $\Sigma_E = \frac{1}{2} \hat{F}$. The predictive Fisher gives the local quadratic approximation under our half-sum KL convention, not generally the observed-label loss Hessian. Empirical and predictive Fisher are distinct [3]; substitution needs validation.

The frozen gate requires relative prediction error at most 0.25 on all three scaled probe directions: random, the last two checkpoints’ update, and the endpoint order gap. Cross-entropy increases are 0.0050, 0.0337, and 0.0087 nats, respectively. Table 2 reports the relative errors. No family passes; the planned projection ladder and Fisher-weighted direction timing are therefore unrun.

Table 2: Local-metric certification on one endpoint pair. Each entry must be ≤ 0.25 ; the predictive-Fisher diagonal uses Monte Carlo sampling. A full empirical-Fisher diagnostic also fails all three probes.

Estimator	Random	Update	Order gap	Passed
Diagonal empirical Fisher	1.11	0.77	0.47	0/3
Per-tensor empirical Gram	1.90	0.61	1.31	0/3
Diagonal predictive Fisher	0.524	0.811	0.126	1/3

The predictive diagonal has 13% error along the endpoint gap, yet underpredicts update-direction KL by about 5.3 times. *Agreement along the selected order-gap direction does not certify a local metric*. The pattern is consistent with missing cross-parameter curvature; it does not establish a unique cause. Independent builds reproduce the order-gap pass/random fail pattern, and the direct divergence readout reproduces to relative error 1.42×10^{-10} . Small loss changes alone do not certify the quadratic regime; finite perturbations and estimator sampling remain limitations. No tolerance was relaxed.

5 Discussion and limitations

Order-only output differences persist, with substantial magnitude before raw parameter-space endpoint-direction alignment. The failed validity checks separate observing a trace from measuring its recoverability. Scalar functional divergence, raw parameter direction, functional geometry, and actual correction are distinct objects; none substitutes automatically for another.

Evidence is limited to one model size, three correlated order-only pairs sharing one initialization, one joint pair, and one training-distribution shard. Initialization-only effects and interactions cannot be isolated; independent-run statistical variation and downstream generalization are unmeasured. No universal irreversibility, successful functional transport, computational lower bound, or validity of arbitrary Fisher models follows. A next experiment should certify a metric with cross-parameter structure before projection/recovery claims, with probe criteria frozen anew. Low-rank proposals must test directions outside their represented span. Functional recoverability and cheap history correction remain open.

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